HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented GPU computing applications using **CUDA** and **OpenCL**, focusing on data parallelism for image processing tasks.
- Optimized and debugged real-time graphics applications by profiling performance bottlenecks, resulting in a **20% reduction** in frame latency.
- Developed and validated embedded software for cloud edge devices, contributing to the design of system-level software solutions.
- Collaborated with cross-functional teams, including systems and hardware engineers, to define and implement software interfaces.
- Applied software engineering principles to modify and validate specialized utility programs for product development lifecycles.
- Participated in code reviews and regression testing to ensure the quality and stability of software releases.
- Authored technical documentation for software projects, detailing design specifications and implementation details.

SKILLS

Programming Languages – C++, CUDA, OpenCL, Python, Java, C#

Graphics & Compute Frameworks – Vulkan, OpenGL, DirectX, Metal, Unity, Unreal Engine, SYCL

Development Tools & Methodologies – Git, Jenkins, Jira, Confluence, Agile, Scrum

Cloud & Embedded Systems – AWS, EC2, S3, Lambda, Docker, Linux

Data Analysis & Visualization – Power BI, Tableau, DAX, R, pandas, NumPy, scikit-learn

PROJECTS

GPU-Accelerated Image Processing Pipeline

- Engineered a Python-based pipeline to ingest and process large image datasets, integrating **CUDA** kernels for parallel computation.
- Optimized image filtering algorithms on the GPU, achieving a **35% speedup** in processing time compared to CPU-bound operations.
- Developed unit tests and integrated the pipeline with a CI/CD system using Jenkins for automated build and deployment.

Real-Time Graphics Rendering Engine Module

- Developed a module for a custom graphics engine, implementing rendering logic using **Vulkan** and optimizing shader performance.
- Debugged and profiled rendering performance using tools like RenderDoc, identifying and resolving bottlenecks that improved frame rates by **~18%**.
- Collaborated with a small team to integrate the module into the main engine codebase, adhering to defined architectural patterns.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)